

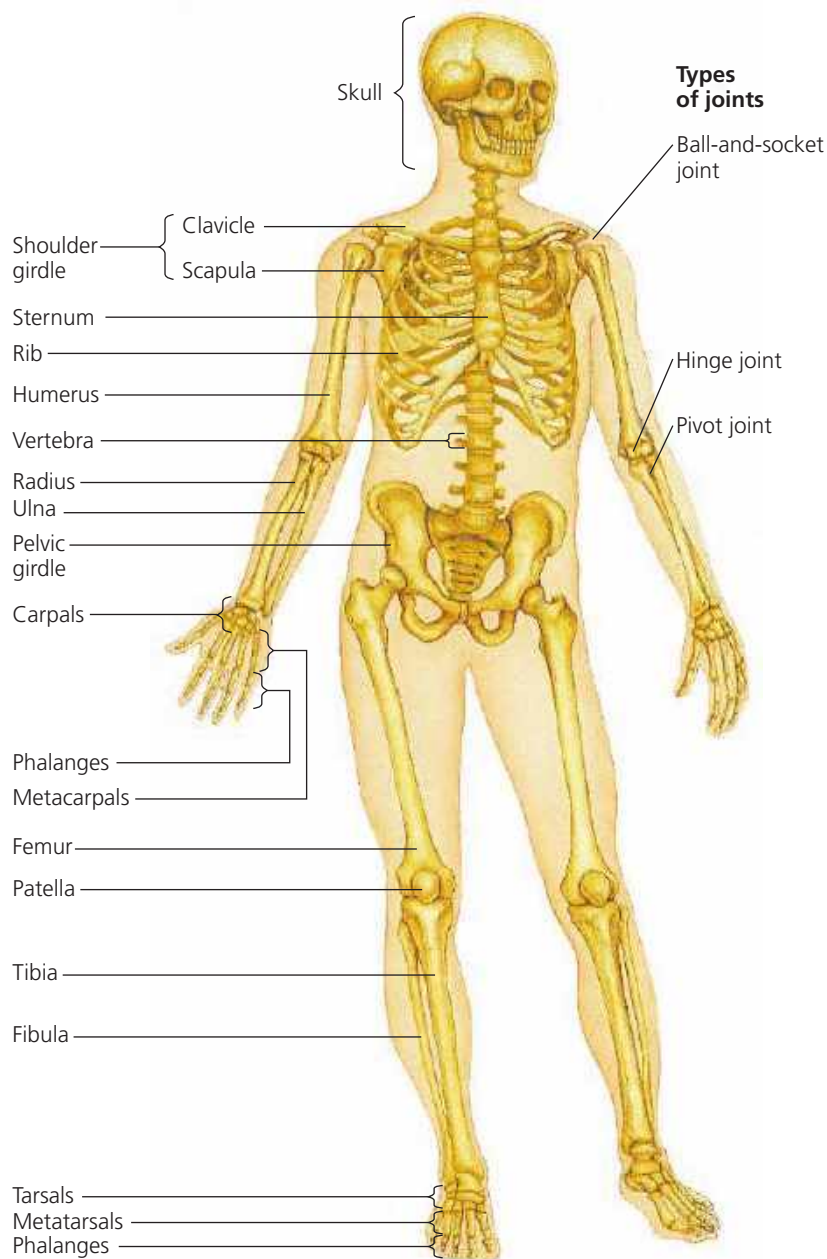
(see Figure 40.5). The mammalian skeleton contains more than 200 bones, some fused together and others connected at joints by ligaments that allow freedom of movement (**Figure 50.37** and **Figure 50.38**). Cells called *osteoblasts* secrete bone matrix and thereby build and repair bone (see Figure 40.5). *Osteoclasts* have an opposite function, resorbing bone components in remodeling of the skeleton.

How thick does an endoskeleton need to be? We can begin to answer this question by applying ideas from civil engineering. The weight of a building increases with the cube of its dimensions. However, the strength of a support depends on its cross-sectional area, which only increases with the square of its diameter. We can thus predict that if we scaled up a

mouse to the size of an elephant, the legs of the giant mouse would be too thin to support its weight. Indeed, the body proportions of large animals are very different from those of small ones.

In applying the building analogy, we might also predict that the size of leg bones should be directly proportional to the strain imposed by body weight. Animal bodies, however, are complex and nonrigid. In supporting body weight, it turns out that body posture—the position of the legs relative to the main body—is more important than leg size, at least in mammals and birds. In addition, muscles and tendons, which hold the legs of large mammals relatively straight and positioned under the body, actually bear most of the load.

▼ **Figure 50.37** Bones and joints of the human skeleton.



▼ **Figure 50.38** Types of joints.

Ball-and-socket joint



Ball-and-socket joints are found where the humerus contacts the shoulder girdle and where the femur contacts the pelvic girdle. These joints enable the arms and legs to rotate and move in several planes.

Hinge joint



Hinge joints, such as between the humerus and the head of the ulna, restrict movement to a single plane.

Pivot joint



Pivot joints enable rotating the forearm at the elbow and turning the head from side to side.